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Molecular-scale mechanics of a crystalline polyethylene interface formed by projection of carbon nanotube structures

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ABSTRACT

The semicrystalline nature of polymer chains induced by the intrusion of carbon nanotubes (CNTs) is a decisive factor in understanding the properties of interfacial regions. Herein, the structural and mechanical behavior of polyethylene (PE) formed around CNT surfaces were determined using a combination of classical molecular dynamics simulations and subcontinuum-scale analysis. The results suggest that inserting CNTs into the polymer domain forces the PE chains to align along the longitudinal axis of the CNTs. The interface functions as a nucleating agent, and the area occupied by the crystalline phase propagates radially as PE chains are stacked over long distances. The crystalline PE, grown with CNT as the central axis, exhibits extreme mechanical anisotropy, particularly for normal and shear deformations in the transverse direction. Therefore, this study provides an important theoretical guideline for understanding polymeric semi-crystals around CNTs.

1. Introduction

Controlling the crystal morphology of polymer systems has long been a fundamental research interest in polymer physics. Material properties vary greatly depending on the conformational difference of the polymer chain and its stacking state [1,2], suggesting that macroscopic material properties are inherently tied to molecular-scale structural properties. Hence, a clear understanding of the correlation between polymer morphology and material properties, including the crystal phase, holds great promise for diverse industrial and academic research fields.

The strategic design of microstructural morphology through precise control of crystallization dynamics is particularly promising as it can substantially improve the thermo-mechanical properties of semi-crystalline polymers. The thermal treatment including annealing, aimed at augmenting the concentration and thickness of crystalline lamellae, have been shown to significantly enhance the elastic modulus and yield strength of materials [3,4]. Such improvement is believed to be mainly due to the extended length of the crystal segments, which effectively suppresses interchain slip during external deformation and improves structural resistance. As an engineering approach, it has been demonstrated that controlling the crystal shape through oscillatory shear injection molding can affect the long-term mechanical stability of the product [5]. The shish-kebab structure and the interconnected lamellae within it contribute to improving dimensional stability by substantially suppressing chain rearrangements. Similarly, some studies have reported that it is possible to achieve high thermo-mechanical properties from polymer to metal level by manipulating the content and orientation of polymer crystalline phases [6–10]. In essence, sophisticated control of crystal morphology appears to be a highly effective strategy for extending the intrinsic properties of polymers.

One intuitive strategy for manipulating the formation behavior of semi-crystalline polymers is using nucleating agents [11,12]. In this regard, numerous studies have been reported employing carbon nanotubes (CNTs) as nucleating agents to induce the crystallization of polymer chains [13–15]. Under low-concentration conditions, CNTs function as heterogeneous nucleation sites by reducing the nucleation potential barrier of the surrounding polymer chains. This is mainly due to the inherent dimensional properties of CNTs composed of sp^2 bonds, and the resulting epitaxial crystallization creates an inherent crystalline structure distinct from the bulk region [16]. In particular, polymers interacting with CNTs get spontaneously oriented along the nanotube axis owing to the extremely high curvature of the CNT cross-section [17,18]. The resulting periodic disc-shaped polymeric crystals, called nanohybrid shish-kebabs, can be utilized for non-covalently functionalizing CNTs or

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promoting their dispersion [19,20].

The above interesting observations prompted an in-depth investigation into the crystal structure of polymers under interaction with one-dimensional nanofiller. Using Monte Carlo simulation, Z. Gu et al. investigated the influence of crystal shape and growing manner according to nanoparticle dimensions [18]. The study revealed that one-dimensional nanoparticles have the strongest ability to induce polymer crystallization compared to lower-dimensional or higher-dimensional nanoparticles and can induce crystal formation with uniform orientation. The mechanisms by which aggregation of nanofiller determines the polymer crystal structure were also comprehensively understood [21]. The regions completely surrounded by multiple nanofillers exhibit accelerated crystal formation and growth compared to the exterior. The polymer chains inside the filler network experience strong structural constraints, which improves segment orientation and reduces structural entropy. Therefore, the chains inside the network participate first in crystal nucleation. Furthermore, the length influence of one-dimensional nanofillers on the crystalline structure was also investigated. The results confirmed that segments of the interfacial region of systems with shorter fillers exhibit lower orientation order parameters, resulting in reduced nucleation capacity [18].

The morphology of CNT-induced semi-crystalline polymers is closely related to the mechanical properties of the materials, including elastic modulus and fracture toughness. In particular, the improvement in the apparent mechanical properties of the material as the polymer crystallization proceeds is highly encouraging [22–24]. This is attributed to the fact that crystallization process induces stronger interfacial bonding with CNTs as well as improves the stiffness of the polymer matrix. Compared with amorphous nanocomposites, crystallized nanocomposites maintain a constant load-transfer efficiency at the interface under mechanical tension by preventing premature fractures in the interfacial region [25]. An experiment that aimed to forcibly control the crystallinity of the polymer on the CNT surface revealed interesting findings. CNTs functionalized with interfacial lamellar showed a higher mechanical performance when composed of nanocomposites than those without, suggesting that interfacial load transfer performance is significantly enhanced by pre-crystallized lamella chains [26].

The aforementioned studies provide empirical evidence that the interactions between CNTs and polymer chains considerably influence the overall properties of the material and its morphology. However, there are only a few theoretical modeling studies that explain the crystallization mechanism of polymers surrounding CNTs and the related changes in the physical properties of the microstructures. The quantification of the mechanical and deformation properties of polymer chains around nanotubes is essential for elaboration of the composites mechanics incorporating CNTs as nanofiller. However, the shapes and properties of the crystalline phases growing around the nucleating agent are distinct due to the physicochemical interactions of the filler and matrix, which is unpredictable using the linear combination of previously known crystalline and amorphous structures of bulk domains.

Motivated by the above, in this study, the structural and mechanical properties of polyethylene (PE) undergoing local crystallization around single-walled CNT surfaces were theoretically evaluated from a nanomechanical perspective. The non-isothermal crystallization process was simulated using all-atom molecular dynamics (MD) simulations, and the changes in the spatial distribution, local arrangement, and interfacial interactions of the crystalline stems depending on the crystallization progress were analyzed. The correlation between the morphological features and the mechanical properties of the microstructure was also further investigated from the mechanical test and the virial stress-based subcontinuum analysis method. Therefore, this study not only clarifies the process of determining the polymer morphology by the insertion of CNTs, but can also serve as a guideline for systematically addressing the design factors of crystalline polymer nanocomposites

2. Modeling of crystallized PE systems around CNTs

The PE/CNT nanocomposite unit cell was modeled using the commercial MD software Materials Studio 2020 (Dassault Systemes, France). The crystallization and characterization of the structural and mechanical properties were performed using the open-source MD code LAMMPS (Sandia Labs, USA).

Fig. 1 shows the components of the unit cell modeled for MD simulation. A PE chain consisted of 500 repeating units, and each CH_2 group comprising the backbone of the chain was represented as a united atom (UA) with a molar mass of 14.02 g/mol. Meanwhile, a single-walled CNT with a length of 59.0 Å and a chiral index (10,10) was also prepared. To model the interfacial region of the PE/CNT nanocomposites, 14 PE chains, and 1 CNT were inserted together in a cuboid simulation box. The PE chains were scattered in the simulation box in random directions and conformations, whereas the CNT was centered in the box along the z-axis. The length of the z-axis of the simulation box was 59.0 Å, which was the same length as that of the CNT, with periodic boundary conditions applied in all three directions. Both ends of the CNT were covalently bonded across the simulation box boundary; therefore, the modeled CNT behaved as if it were of infinite length. To avoid undesirable interference between the PE chains inserted at random positions, the x- and y-directions of the simulation box were set sufficiently large such that the entire system had a very low initial density of 0.1 g/cc.

The DREIDING force field was used to describe all interactions between the constituent atoms [27]. It has been demonstrated that the DREIDING force field can provide reliable information on the crystallization temperature [28], crystal structure [29,30], and thermal/-mechanical properties [31,32] of PE, and also successfully predict the extremely high axial modulus of CNTs [33,34]. In particular, the non-covalent interaction behavior between the CNT and PE described using DREIDING is in good agreement with the calculations obtained from the ab initio studies [35]. After the initial placement of the molecules in the simulation box, the internal potential energy of the entire system was minimized using the conjugated gradient method with an energy cutoff of 10 kcal/mol. Subsequently, the density of the system was gradually increased through time integration for 5 ns under an isothermal–isobaric ensemble (NPT) at 300 K and 1 atm. The motions of all constituent atoms were calculated using the Verlet algorithm with a time step of 0.5 fs.

The crystallization of PE chains around the CNT was achieved according to the following procedure: first, the PE/CNT nanocomposites were exposed to an extremely high temperature of 600 K for 5 ns in the NPT ensemble. Because 600 K far exceeds the crystallization point calculated from the DREIDING force field (400 K, [28]) and the experimental melting point (398− 406 K, [36,37]), the PE chains become completely amorphous. In the process, temperature, applied pressure, and system density all reached equilibrium (Fig. S1). The temperature of the system was then decreased at a constant rate of 3 K/ns under the NPT ensemble until it reached 300 K.

During cooling the system, the change in the mass crystallinity of PE (X_c) was monitored using the method described by Moyassaria et al. [38,

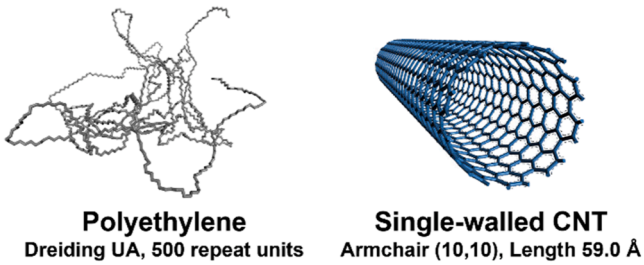


Fig. 1. Molecular structures of the PE chain (left) and CNT (right) modeled in this study.

39]. This criterion defines the stem of crystal through molecular-scale linearity of the PE chain. First, for each bead i that constituted the PE chain, a direction vector (formed with an even number of neighboring beads) was calculated. Beads that were up to six spaces away from the reference bead were considered in calculating the relative position; therefore, a total of six direction vectors were defined for each bead (i.e., $\vec{V}_{i-2,\,i}$, $\vec{V}_{i,i+2}$, $\vec{V}_{i-4,i}$, $\vec{V}_{i,i+4}$, $\vec{V}_{i-6,i}$, and $\vec{V}_{i,i+6}$, also see Fig. S2). The three angles were then calculated from the six direction vectors as follows: $\theta_1 = \cos^{-1}\frac{\vec{V}_{i-2,\,i}\cdot\vec{V}_{i,i+2}}{|\vec{V}_{i-2,\,i}||\vec{V}_{i,i+2}|}$, $\theta_2 = \cos^{-1}\frac{\vec{V}_{i-4,\,i}\cdot\vec{V}_{i,i+4}}{|\vec{V}_{i-4,\,i}||\vec{V}_{i,i+4}|}$, and $\theta_3 = \cos^{-1}\frac{\vec{V}_{i-6,\,i}\cdot\vec{V}_{i,i+6}}{|\vec{V}_{i-6,\,i}||\vec{V}_{i,i+6}|}$. When θ_1, θ_2, and θ_3 are all lower than 15°, the bead is considered to be a stem constituting a crystal. In other words, the minimum unit of a stem is defined as 13 carbon atoms arranged in a row (about 1.6 nm in length). The mass crystallinity is obtained as the ratio of stem beads to the total beads that constitute the PE chain: $X_c = N_{stem}/N_{PE}$.

The changes in mass crystallinity and system density during cooling are shown in Fig. 2(a). Based on the change in slope of the profile (see Fig. S3), the mass crystallinity exhibits three distinguishable phases depending on the applied temperature: (i) in the high-temperature region, the amorphous nature of the PE chains is maintained because excessive thermal energy provides high polymer mobility; (ii) the potential energy between the beads in the system becomes relatively dominant as the magnitude of the kinetic energy provided by the thermodynamic ensemble decreases. Therefore, the structure of the PE chain is partially crystallized, and the X_c value increases rapidly as it exceeds a certain temperature. The characteristic is also confirmed by the formation of inflections in the change profile of potential energy in the temperature range at which crystallization begins (Fig. S4); (iii) as the crystallites restrict further growth from each other, the crystallinity of the PE converges. Because the crystalline region is more ordered and condensed than the amorphous region, the density profile of the overall system exhibits the same trend as the crystallinity. The nonlinear growth of crystallites during non-isothermal crystallization is consistent with previous experimental and theoretical studies [28,40,41].

Notably, a significant amount of amorphous segments (approximately 40 %) were still present inside the PE chains even after the cooling process was complete; This implies that the PE chains inevitably contain amorphous defects inside due to irregular growth rate and mutual interference between crystalline domains (see **Amorphous defects after crystallization** section and Fig. S5 of Supplementary Data). The process and principle by which the interaction with the CNT surface spatially redistributes the amorphous defects of the PE chains is also within the scope of interest. Such properties are particularly important for understanding the effect of external loading on the stress distribution of the PE/CNT nanocomposites, which is discussed in detail in Section 3.2.

Five PE/CNT models with different X_c ranging from 15 to 60 % were selected to characterize the structural and mechanical properties of the microstructures depending on their crystallinity. The snapshots of the equilibrated models for each mass crystallinity are shown in Fig. 2(b). The detailed structural information, including the unit cell length, density, and CNT volume fraction of the models, is summarized in Table 1. Also note that the structures and properties of crystallized PE modeled in a given environment are in good agreement with those reported experimentally, both phenomenologically and quantitatively, as shown in Table S1.

3. Results and discussion

3.1. Morphology of PE chains crystallized on the CNT surface

During the non-isothermal crystallization, the crystalline PE around the CNT surface exhibits several interesting morphological features (Fig. 3(a)): first, a microstructure with highly ordered PE chains was observed intensively around the CNTs. The structural differences between the CNT surface periphery and far-field were particularly evident under low-crystallinity level. This finding indicates that the CNT acted as nucleation sites for crystallization, producing anchoring forces that suppressed the diffusion of neighboring polymer chains; this is in agreement with the experimental observations that CNTs promote the growth of polymer crystallites in the lateral direction of the nanotubes [42]. Second, after sufficient crystallization was completed, defective regions composed of amorphous segments still remain in the PE matrix, especially away from the CNT. Finally, the most interesting point is that the crystallized PE at the interface formed a periodic cylindrical shell structure concentric with the CNT. This unique morphology can only be exhibited by PE chains crystallized around CNTs and is completely different from linear crystal structures found in the bulk PE domain.

The adsorption behavior of the PE chains by interaction with the CNTs was quantitatively evaluated from the density distribution along the radial distance from the nanotube surface (Fig. 3(b)). The density distribution of the microstructure was derived by setting up a series of cylindrical layers with a thickness of 0.2 Å concentric with the CNT, and calculating the density. The interaction force with the CNT surface forms a densified polymer layer in the immediate area. Subsequently, as the radial distance increased, the density gradually converged to the bulk value with significant oscillations. The morphological features became more prominent in microstructures with high crystallinity. The value of the first peak in the distribution increased from 4.57 to 5.42 g/cc as the crystallinity increased from 15 to 60 %. The point at which the oscillations of the profile vanish has also moved further away, from 15 Å to 30 Å. Meanwhile, the converged density of PE matrix increased significantly from 0.68 to 0.92 g/cc due to the highly aligned and condensed

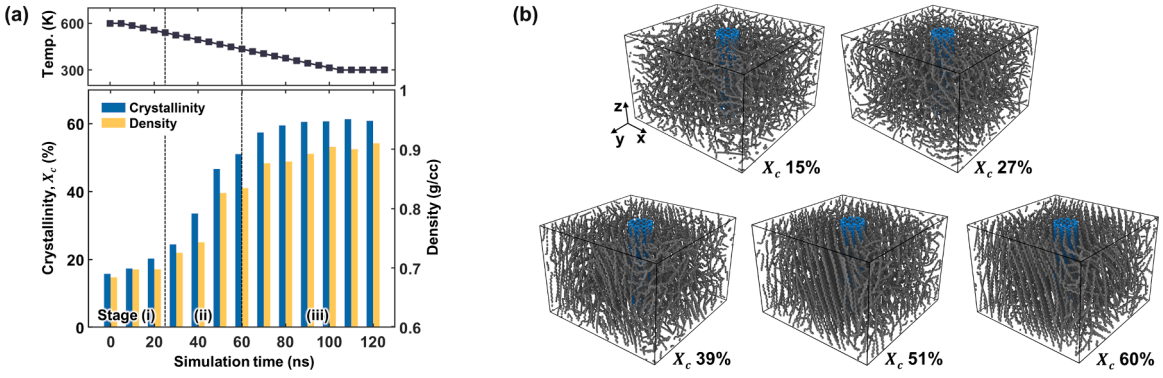


Fig. 2. Crystallization of PE chains around CNT surface. (a) Changes in density and mass crystallinity during cooling-down simulation. (b) Snapshots of the PE/CNT nanocomposite models extracted according to the level of mass crystallinity.

Table 1
Structural details of PE/CNT models with increasing crystallinity.

Model	Number of molecules PE	CNT	Mass crystallinity (%)	Unit cell length (Å^3)	Density (g/cc)	CNT volume fraction (%)	Orientational order
Xc 15 %	14	1	14.9	92.8 × 92.8 × 58.5	0.684	1.67	0.154
Xc 27 %			27.3	88.4 × 88.4 × 58.4	0.754	1.84	0.225
Xc 39 %			39.2	85.9 × 85.9 × 58.5	0.798	1.94	0.566
Xc 51 %			50.8	84.1 × 84.1 × 58.3	0.834	2.04	0.657
Xc 60 %			60.2	80.2 × 80.2 × 58.5	0.916	2.24	0.704

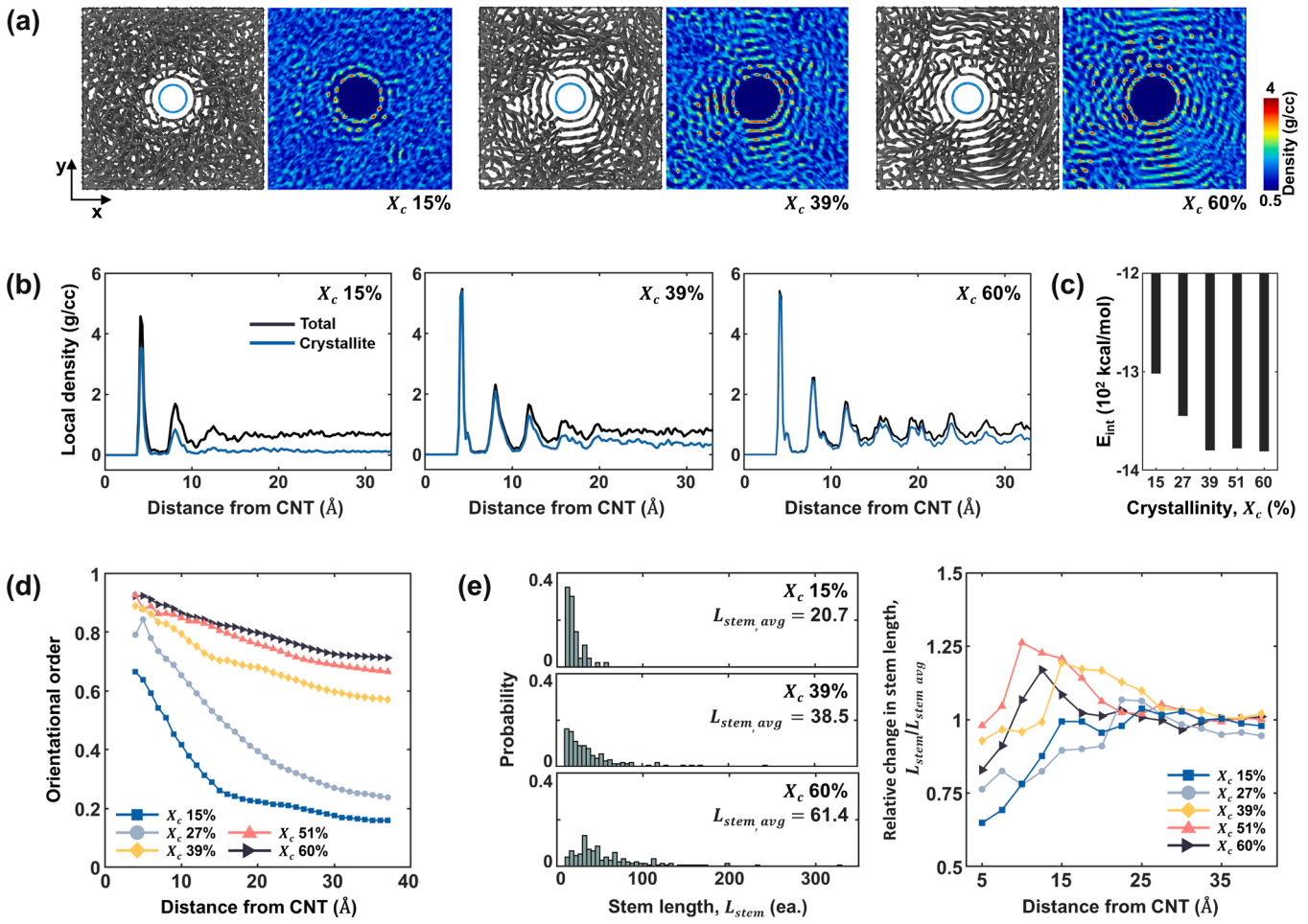


Fig. 3. Structural analysis of PE chains crystallized on the CNT surface. (a) Molecular configurations and spatial density distributions of PE/CNT according to crystallinity. The spatial density colormap of the PE matrix domain for each case was calculated by dividing the simulation box evenly into 75 × 75 squares along the *x*- and *y*-axes. (b) Density distributions of PE chains along the radial distance from the CNT surface. The distribution of all PE chains and PE components with crystallinity are shown separately. (c) Interfacial interaction energy between the PE chains and the CNT. (d) Changes in the orientation order of PE chains with radial distance from the CNT surface. (e) Stem lengths of PE chains based on the crystallinity (left) and spatial distribution of stem lengths according to the radial distance from the CNT surface (right). The spatial distribution profile was normalized to the average stem length value observed at each crystallinity condition.

structure of crystalline domains.

The density distribution of crystalline PE alone, which comprises the stem, suggest a more interesting finding. The contribution of crystalline PE to the density profile from the CNT exhibits a high spatial bias. Specifically, the density profile of the regions close to the CNT surface was dominated by well-ordered crystalline segments. This finding is consistent with the configuration analysis showing that CNTs function as nucleation seeds for the crystallization of polymer chains. In addition, because the anchoring force generated by the CNT is inversely proportional to the distance from the nanotube surface, the mobility of the distantly located PE remained high, even during crystallization of the interfacial region. This enables the free growth of crystallites without spatial constraints on the nanotube surface. and consequently, crystallites in the interfacial region exhibit a higher structural order than those in the bulk region. Even when the crystallinity was sufficiently high (e.g., X_c 60 % model), the difference between the two density profiles was clearly distinct in the region far from the CNT.

The interfacial interaction energy (E_{int}) was calculated to investigate the change in the bonding state between the CNT and PE chains as crystallization progressed. First, the total potential energy of the prepared PE/CNT microstructure (E_{system}) was calculated. Next, we deleted the CNT component in the microstructure and calculated the potential energy formed by the PE matrix alone (E_{PE}). During the calculation process, the force field parameters, atomic polarizations, and atomic positions of the PE matrix were all preserved. Similarly, the potential energy was also calculated when all PE chains in the microstructure

were deleted, leaving only the CNT (E_{CNT}). Finally, the nonbonding energy formed between the PE matrix and the CNT filler can be derived by substituting the potential energies obtained for each of the three systems into the following equation:

$$E_{int} = E_{system} - E_{PE} - E_{CNT} \quad (1)$$

The results show that PE with higher crystallinity has stronger bonding with the CNTs (Fig. 3(c)). This is attributed to the fact that when crystallization was initiated, the nanosized voids remaining between the entangled polymer chains were removed, and the space was filled with condensed microstructures. In the CNT/PE system considered in this study, the CNT served as the crystallization seeds, so the free volume in the interfacial region was rapidly removed even under low-crystallinity conditions (Fig. S6). In addition, because the crystallization preemptively started at the CNT surface and grew into the bulk region, the interfacial interaction energy rapidly converged after being saturated in the X_c 39 % model.

The change in the alignment properties of PE chains by CNT insertion and crystallization was evaluated. The direction vector penetrating the PE beads two places away from the reference bead was considered ($\vec{V}_{i-2,\ i+2}$), and the angle (θ) made with the nanotube axis was measured. The measured angular distribution was quantified as a representative scalar value using the Herman's orientational order parameter, defined as

$$S = \left\langle \frac{3cos^2\theta - 1}{2} \right\rangle \quad (2)$$

where the bracket symbol indicates the average of all cases within the region of interest. Because the main concern was the change in the structural properties with distance from the nanotube surface, the value of S was calculated in the space of the cylindrical shell surrounding the CNT. The space was considered cumulatively, increasing by 1 Å until reaching the boundary of the simulation box.

The results showed that CNT induced the co-alignment of adjacent PE chains along the length direction, leading to a high order of magnitude on the nanotube surface (Fig. 3(d)) because in the trans state, where the internal potential energy of PE is minimized, the molecular chain has high straightness and resistance to bending deformation. Particularly, from the viewpoint of minimizing the strain energy, the PE chains prefer to align with the nanotube rather than wrap around the nanotube surface. The ordering effect was initially prominent at approximately 12 Å, corresponding to the van der Waals distance with the CNT, but was propagated over long distances during crystallization. In particular, the PE chains of the X_c 60 % model had a high directionality of $S > 0.7$ at all distances because the entire nanocomposite matrix was composed of single crystallites.

The stem length distribution of the PE during the crystallization reaction was analyzed (the left of Fig. 3(e)). The stem-length histogram exhibited a unimodal distribution, which was consistent with previous theoretical findings [38]. As the crystallization progressed, the average number of atoms constituting the crystallite stem increased from 20.7 to 61.4 ea., which indicated thickening of the crystallites owing to the anchoring and diffusion of amorphous segments on the crystallite surface. Moreover, the variance in the histogram profiles increased with crystallization, therefore suggesting that the matrix domain was filled with PE stems of various lengths.

The relationship between the interaction with the CNTs and the crystal growth properties was elucidated by analyzing the stem length distribution from the nanotube surface (right side of Fig. 3(e)). Notably, the profile shows three distinct stages based on the relative distance from the CNT filler. At a distance of ~10 Å, where the interaction force of the CNT was prominent, the stem length of the PE was shorter than that of the bulk region. This finding is attributed to the CNT exerting an anchoring force on the surrounding polymer chains, which significantly restricts diffusion and extension of the crystallite segments. However, at a distance of ~15 Å, the trend was reversed, resulting in an enlarged length compared to the bulk region. Although the non-covalent forces generated by the CNTs disappeared, the direction of crystallization were still provided by the previously crystallized segments, leading to a high growth rate of the stems. Finally, as the distance increased further, the stem length converged to the bulk value.

The implications of the preceding structural analysis are straightforward. During non-isothermal crystallization, CNTs serve as nucleation seeds for crystallization by forcing PE chains to their surface. Because interaction between heterogeneous materials dominate crystallization behavior, the morphology of the interfacial crystals involves the geometrical feature of the fillers. These features of the interfacial crystals are clearly distinct from that of neat polyethylene, which exhibit a simple orthorhombic structure. Previous experimental and computational studies have demonstrated that the diameter, thickness, and periodicity of interfacial crystals can be influenced by crystallization conditions or polymer composition [42–44]. It is particularly important to report that under most of the conditions considered, the crystallization progress is dominated by the interaction of the nanofillers in the interfacial region. Therefore, the next aspect of interest is to elucidate how the crystalline structure, influenced by adjacent nanofillers, determines the microscopic interfacial properties of the nanocomposites. In the following sections, we explicitly characterize the mechanical properties exhibited by the polymer crystal structure as influenced by the geometry of the filler. By applying transverse normal and transverse shear deformations along the CNT axis, the mechanical response of the oriented polymer chains surrounding the CNT was quantitatively evaluated.

3.2. Mechanical behavior of PE chains crystallized on the CNT surface

The effect of the crystalline structure on the mechanical properteis of nanocomposites was evaluated. In all atomic MD simulations, the atomic stress tensor was computed based on the virial theorem defined by Heyes [45]

$$\boldsymbol{\sigma} = \frac{1}{V}\left(-\sum_{i}^{N} m_i\left(\boldsymbol{v}_i\boldsymbol{v}_i^T\right) + \frac{1}{2}\sum_{i}^{N}\sum_{j\neq i}^{N} \boldsymbol{r}_{ij}\boldsymbol{F}_{ij} \right) \quad (3)$$

where V is the volume, N is the number of constituent atoms, m_i and $\boldsymbol{v}_i$ are the mass and velocity of i th atom, respectively, and $\boldsymbol{r}_{ij}$ and $\boldsymbol{F}_{ij}$ indicate the relative position and interaction force between i th and j-th atoms, respectively. In other words, virial theory characterizes the stress of individual atoms under a given force field, implicitly taking into account both the covalent and non-covalent properties of the modeled polymer chain. In this study, a crystallization simulation was performed under non-isothermal conditions, where the temperature of the system decreased linearly. Therefore, the mechanical properties of PE/CNT at each crystallinity are influenced by both the polymer morphology and kinetic energy applied to the system. In this section, we focused on the structural properties of PE chains determined by the crystallization process from a static perspective. Therefore, all of the considered models were relaxed under the NVT ensemble at 300 K for 200 ps, and mechanical tests were performed under cryogenic conditions (0.1 K) with a constant strain rate of 10^6 /s to suppress the further crystallite growth during deformation.

It is worth to recalling that the CNT implemented in the simulation box are interconnected across the periodic boundary, leading them to behave as cylinders with infinite length. Therefore, the mechanical behavior along the axial direction only provides a trivial solution dominated by the direct stretching of the CNTs. However, as shown in Fig. 3(a), the crystalline PE on the CNT surface formed a cylindrical secondary structure concentric with the intercalated CNTs, which potentially played a crucial role under deformation in the lateral directions. Based on these observations, the interfacial structure was characterized with an emphasis on transverse normal and transverse

shear deformations. Because all PE/CNT models exhibited transversely isotropic behavior, the atomic stress along the transverse normal (σ_{TN}) was calculated by averaging the components of the axes parallel to the stretching direction (i.e., σ_{xx}under x-direction tension and σ_{yy}under y-direction tension). The maximum tensile strain was 0.2 % belonging to elastic region. For the transverse shear stress (σ_{TS}), simple shear strains were applied up to 0.4 and –0.4 %, respectively, and the corresponding stress components (i.e., σ_{xy} under positive shear and σ_{-xy} under negative shear) were averaged.

The atomic stress calculated using the virial theorem provides a quantitative understanding of the effective contribution of each molecule and its sub-elements to the system under external loading [46,47]. While increasing the internal energy of the entire microstructure via forced displacement, the atomic stress can be selectively calculated only for the target of interest. The effective volume occupied by the target of interest within the microstructure was quantified using Voronoi tesselation. From this method, the effective strain energy of the CNT and PE chains can be quantified within the microstructure, and the local distribution of strain energy in the interfacial region can be understood [48–50]. Specifically, the effective stress tensor of subdomain $k$was derived from the following equation:

$$\boldsymbol{\sigma}^k = \frac{1}{V^k}\sum_{i=1}^{N} V\boldsymbol{\sigma}^i \tag{4}$$

where N is the number of atoms contained in subdomain k, V^k is the corresponding volume defined by the Voronoi tesselation, and $\boldsymbol{\sigma}^i$ is the stress tensor of i th atom belonging to the subdomain k. The substantial work of each subdomain under mechanical deformation was quantified using the effective strain energy, as follows:

$$u_{TN} = \int_0^{\varepsilon_0} \sigma_{TN}^k d\varepsilon \text{ and } u_{TS} = \int_0^{\gamma_0/2} \sigma_{TS}^k d\gamma \tag{5}$$

where σ_{TN}^k and σ_{TS}^k are the atomic stress of the subdomain k under the transverse normal and shear loading, and ε_0 and $\gamma_0$are the maximum normal strain and simple shear strain applied to the unit cell, respectively.

The elastic behavior of the system under mechanical deformation were clearly distinguished according to the progress of crystallization in the interfacial region. In particular, completely different responses were obtained in the transverse normal and shear load directions (Fig. 4(a)). Under transverse normal loading, the slope of the stress–strain profile of the PE/CNT interface increased monotonically with increasing crystallinity. The non-covalent bonds between the PE chains were significantly strengthened by the regularly ordered microstructure, which resulted in high structural resistance to deformation. By contrast, no enhancement was observed under transverse shear loading with increasing crystallinity. The PE structure in the cylindrically stacked interfacial region mainly relied on non-covalent bonding forces to achieve load bearing under in-plane deformation and was therefore prone to slipping. In other

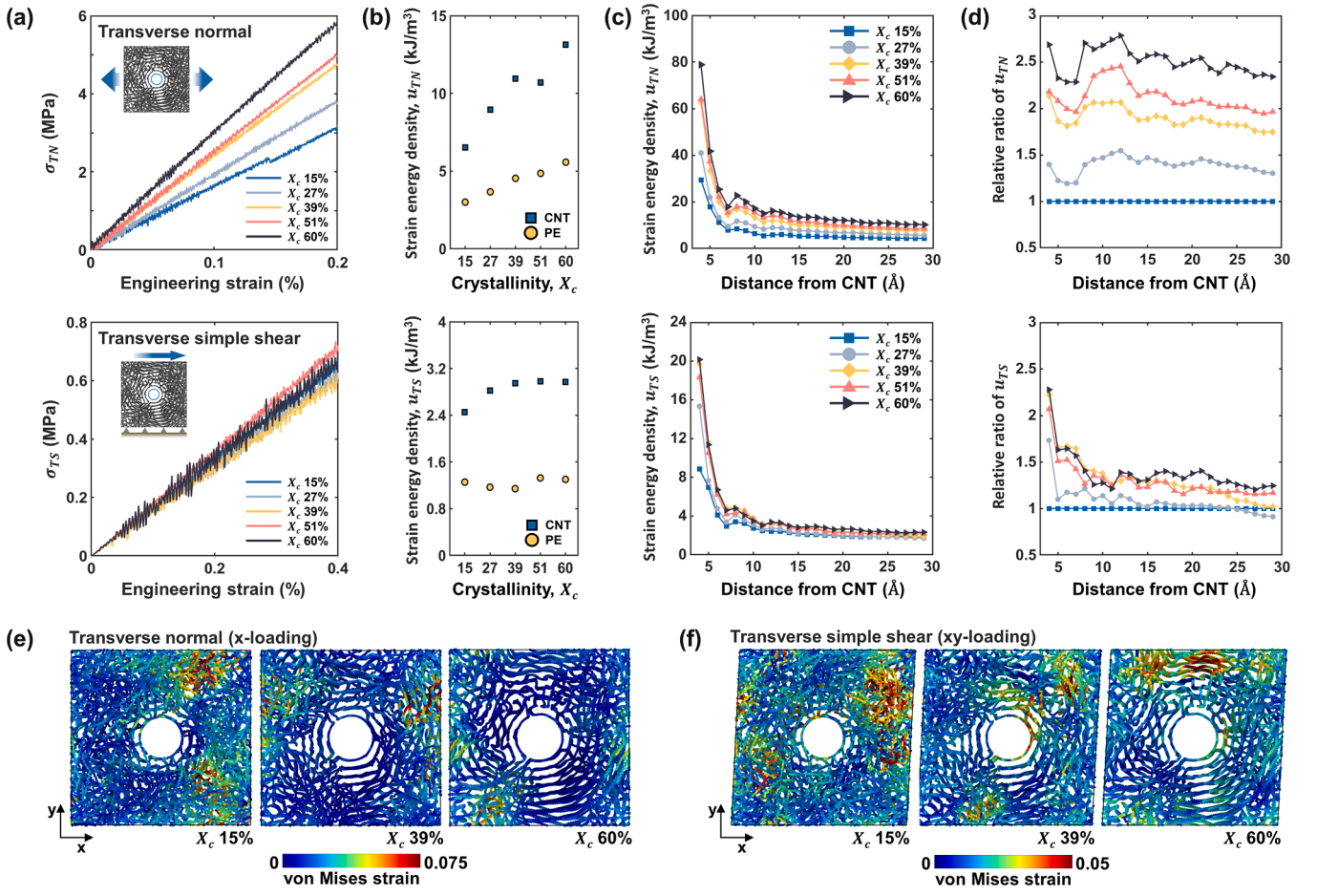


Fig. 4. Mechanical properties of PE/CNT models with different crystallinities. (a) Stress–strain curves under transverse normal (top) and transverse shear loading (bottom). (b) Elastic strain energy densities by molecular component under transverse normal (top) and transverse shear loading (bottom). (c) Distributions of strain energy density of PE matrix around the CNT surface under 0.2 % normal strain (top) and 0.4 % shear strain (bottom). (d) The post-processing of (c) for stressing the relative change in strain energy density distribution compared to the lowest crystallinity condition (i.e., X_c 15 % model). (e) The von Mises strain distributions of PE matrix under transverse normal and (f) transverse shear loading with crystallinity.

words, the stacked PE chains easily distorted the cylindrical shape without changing the interlayer distance under shear deformation, thereby maintaining the system at a low level of strain energy.

To determine the contribution of the components within the system to the formation of mechanical stress, the elastic strain energy was classified for the area occupied by the PE and CNT using Eq. (4) (Fig. 4(b)). Under both deformation conditions, the strain energy density of the CNT was always higher than that of the PE chains. This indicated that the structure of the highly ordered PE chains efficiently transferred the load towards the CNT in the interfacial region. Because crystalline PE chains preferentially grow on the surface of the CNT, the observation is made clearly even under low-crystallinity conditions. However, in transverse normal strain, the strain energy density value of the CNT continued to increase along with the crystallinity of the PE chain, whereas in transverse shear, it quickly converged to a specific value. In addition, the strain energy density of the PE chains shows a high anisotropy depending on the loading direction (i.e. monotonic increase under the transverse normal and convergence under the transverse shear), therefore verifying the morphological characteristics of the cylindrically layered PE chains along the CNT axis.

The origin of the different strain energy density properties depending on the mechanical strain mode was further investigated through the spatial energy distribution of the PE chains on the nanotube surface (Fig. 4(c)). The spatial distributions were obtained by considering several cylindrical layers with a thickness increment of 1 Å from the CNT surface and calculating the strain energy density of the PE chains belonging to each region. To better understand the spatial distribution of the strain energy as crystallization progressed, the normalized results with the profile of the X_c 15 % model are also presented (Fig. 4(d)).

When the PE/CNT was subjected to a transverse normal load, the strain energy density profile was higher at a close distance. The observation is particularly prominent at a distance of ~6 Å, wherein a condensed polymer layer due to the interaction with the CNT appears, indicating that the crystalline PE exhibits higher structural order especially in the interfacial region. This result is consistent with previous structural analysis results showing that highly deformable amorphous segments are preferentially annihilated around the CNT during crystallization (Fig. 3(b)). The crystallized PE chains in the region immediately adjacent to the CNT maintained strong interactions with the CNT while transferring the transverse normal load between the interfaces, generating a remarkable strain energy density. As the crystallinity of the entire system increased, the strain energy density profile exhibited a monotonic upward trend; that is, the rate of increase in the strain energy with crystallinity was spatially uniform within the PE matrix. This outcome suggests that the mechanical properties owing to the molecular-scale structural arrangement of the crystallized PE chains are propagated over a broad domain of the microstructure.

When PE/CNT was deformed under transverse shear loading, the strain energy density had the highest value around the CNT surface, similar to that of a transverse normal load. However, the increase in strain energy with crystallinity was distinctly concentrated only at the CNT periphery without spanning the entire PE matrix area. This observation is completely different from the behavior of transverse normal loading, in which the strain energy distribution monotonically increases with crystallinity. The PE chains wound in a cylindrical shape along the longitudinal axis of the CNT release the increase in internal energy caused by shear deformation, mainly through interlayer slip. The mechanical resistance of the crystallized PE chains to the sliding mode was very low; therefore, it could not be improved by increasing the overall crystallinity of the microstructure.

Finally, the deformation response of crystalline PE under different mechanical loadings was visualized using finite strain theory. The strain tensor of the atoms constituting the PE matrix is defined based on the Lagrangian strain tensor ($\boldsymbol{E}$) as

$$\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) \tag{6}$$

where $\boldsymbol{C}$ and $\boldsymbol{I}$ indicate the Cauchy–Green deformation tensor and the identity matrix, respectively. The strain tensor was calculated from the change in the relative distance between the constituent atoms during mechanical strain. Rigid body displacements such as translation or rotation of the PE chain, were excluded from the calculations, and only the effects due to substantial deformation of the crystalline structure were considered. The mechanical strain experienced by each atom was quantified as a representative value by using the von Mises strain (ε_{vM}) criterion.

$$\varepsilon_{vM} = \frac{1}{3}\left[2\left((\varepsilon_1 - \varepsilon_2)^2 + (\varepsilon_2 - \varepsilon_3)^2 + (\varepsilon_3 - \varepsilon_1)^2\right)^{1/2}\right] \tag{7}$$

where ε_1, ε_2, and ε_3 are the principal strains of $\boldsymbol{E}$. To further highlight the influence of mechanical loading, the PE/CNT unit cells were further deformed up to 5 % under same tensile condition. The stress–strain profiles of the considered models exhibited good linearity in this strain range for both transverse normal and shear loading, so that the PE/CNTs experienced only elastic deformation (Fig. S7).

When the system was subjected to transverse normal loading, the von Mises strain in the PE matrix was significantly reduced as crystallization proceeded (Fig. 4(e)). Such strain suppression is particularly observed in the crystal domains that were initiated from the CNT surface and propagated to the periphery. Therefore, the PE chains have a relatively high atomic strain only in the noncrystalline defect region. However, no significant difference was observed in the strain field with respect to the crystallinity of the PE matrix under a transverse shear load (Fig. 4(f)). The crystalline domain exhibited a level of deformability similar to that of the amorphous state, which is attributed to the structural fragility of the chains arranged concentrically along the nanotube axis against shear loading.

From the above observations, the main morphological features of PE crystallized around the CNTs were identified. The PE chain surrounding the CNT in a cylindrical shape served as an initial seed for the crystalline phase, and the crystalline-phase region expanded as its morphological characteristics were repeatedly transferred to adjacent PE chains over a long distance. Moreover, the PE molecules were highly aligned along the nanotube axis, forming strong non-covalent bonds with adjacent chains. From a structural mechanics perspective, this bonding relationship results in a particularly high load transfer effect for deformation modes that widen the interlayer distance between the stacked PEs. Therefore, crystalline PE chains having concentric axes with CNTs can be inferred to be particularly advantageous for transverse normal loads, and the results of mechanical test simulations of microstructures strongly support this. The observation that the crystalline PE chains exhibit a low level of mechanical stiffness, similar to that of the amorphous state under transverse shear loading, agrees well with their morphological characteristics.

4. Conclusion

In this study, the interfacial properties of PE crystallized on the CNT surface were characterized from an all-atom computational perspective. Theoretically, PE chains crystallize along the unique cylindrical shape of the CNTs, and their morphology determines the anisotropic properties of the interfacial layer in the cross-sectional direction.

One of the key observations was that the surface structure of the CNT functioned as a nucleation seed for crystallization by anchoring adjacent PE chains through strong non-covalent forces. The nucleation seeds initially generated on the CNT surface gradually expand the region of crystalline PE radially while the microstructure undergoes cooling. In particular, as the crystallization proceeded, the PE chains spontaneously aligned along the length of the nanotube, significantly increasing the

orientation order of the entire microstructure. The crystallization process also reduces the nanosized pores distributed between the entangled PE chains, making adsorption on the CNT surface more robust. The structural characteristics of the crystallized PE chains formed on the surface of the nanotubes induced outward rearrangement of the adjacent PE chains; therefore, the anisotropic nature of the structure became more pronounced as crystallization proceeded.

The structural anisotropy of the microstructure was analyzed in depth from the mechanical response using transverse normal and transverse shear deformation tests. The cylindrically crystallized PE structure along the CNT geometry was particularly resistant to deformation, which increased the relative distance between the stacked layers. Therefore, the mechanical response of the microstructure under transverse normal loading improves monotonically with increasing crystallinity. Conversely, transverse shear loading only induces interlayer sliding without damaging the non-covalent structure between the crystallized PE layers, and the associated mechanical properties remain low, regardless of crystallinity. The anisotropic responses were further characterized in terms of the subcontinuum through the strain energy density of the subdomain and von Mises strain distribution.

The mechanism described herein provides intuitive and important molecular-level insights into the field of soft matter engineering utilizing CNT-modulated polymer crystallization. These findings provide theoretical guidelines for the robust design of nanohybrid materials with precisely controlled crystal morphologies. This work will be further extended in the near future to the design of other nanoscale parameters that have important effects on crystallization kinetics. Specifically, the effects of physicochemical properties, such as shape [18], content [51], dimension [51], and surface treatment [52] of various nanofillers, on the crystallinity of polymers placed at the interface are in progress.

CRediT authorship contribution statement

Hongdeok Kim: Methodology, Software, Writing – original draft. **Joonmyung Choi:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

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